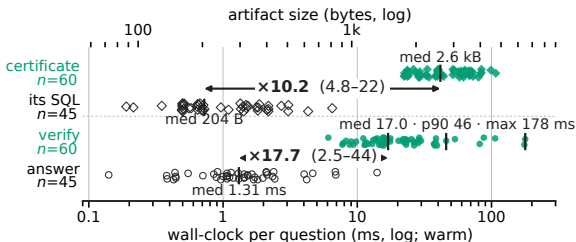


(a) cost: the certified path loses on both axes



warm tail: CODE-Q1/CODE-Q2 (0.18/0.18 s) replay the disclosure-cell census on the largest self-join; the full-history escape class (Prop 5.11) has no instance on this base

(b) ablation ladder: every rung recomputed (n = 60)

